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BAGONG PILIPINAS

CC 101 – Introduction to Computing

Laboratory Activity No. 1 | Week 2: Introduction to Computing (CO1) | Unit 2, Computing Career Pathways

Explore Career Pathways

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Course/Program: BSCS 1-1 **Date: Sept 16, 2026,**

I. Objectives

At the end of this activity, you should be able to:

- Identify the major computing career pathways introduced in lecture and describe what each one involves.
- Choose one computing career pathway that best matches your interests and explain why.
- Map a realistic entry-to-expert roadmap for your chosen pathway, including representative job titles and key skills at each stage.
- Compare your current skills against your chosen pathway's requirements and identify a concrete plan to close the gap.

II. Materials Needed

- Your CC 101 Unit 2 lecture slides/notes on Computing Career Pathways
- This worksheet (printed copy, or the digital file to fill out and submit)
- Optional: internet access, for Part E (real job posting search)

III. Background

Computing is not just "programming." As introduced in the Unit 2 lecture, the field branches into several distinct career pathways, each with its own operational scope, required technical competencies, representative job titles, and professional standard or reference body: Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, Cybersecurity & Threat Defense, and Information Systems & Enterprise IT. Each pathway also has its own career roadmap — the entry-level, mid-level, senior-level, and expert/leadership roles a professional typically grows through. This activity asks you to review those five pathways, choose the one that interests you most, and map out what growing in that pathway could look like for you.

IV. Procedure

Part A – Recall the Five Pathways

1. Using your lecture notes or slides, complete Table 1 by writing the core operational scope and one to two representative job titles for each of the five computing career pathways.

Part B – Choose Your Pathway

2. From the five pathways, choose ONE that interests you the most.
3. In two to three sentences, explain why you chose it (e.g., skills you already enjoy, a subject you like, a job title that excites you).

Part C – Map Your Career Roadmap

4. If your chosen pathway is Software Engineering, Systems & Cloud Infrastructure, Data Science & AI Engineering, or Cybersecurity & Threat Defense, use the career roadmap shown for it in the lecture. If

you chose Information Systems & Enterprise IT (no roadmap was shown in lecture), research typical entry-level, mid-level, senior-level, and expert/leadership roles for this pathway online.

5. Complete Table 2 with example job titles and key skills/responsibilities at each of the four career stages.

Part D – Skill Gap Self-Assessment

6. Looking at the essential technical competencies for your chosen pathway (Table 1 / lecture), list at least five in Table 3. For each, mark whether you already have some experience with it, and briefly note how you plan to learn it if you don't.

Part E – Real-World Job Search (Optional / Bonus)

7. Search a real job platform (e.g., LinkedIn, Indeed, JobStreet, Kalibrr) for one entry-level job posting that matches your chosen pathway. Record it in Table 4.

V. Data and Findings

Table 1. The Five Computing Career Pathways (Recap — Part A)

Pathway	Core Operational Scope	Representative Job Title(s)
Software Engineering	It focusses on planning, checking, and building apps and software that run on rules and calculations.	Software Engineer, Backend Developer, Full-stack Engineer.
Systems & Cloud Infrastructure	Look after physical and online computer gear, set up tools, keep things running smoothly, and make new versions available.	Systems Administrator, devOps engineer, Site reliability Engineer (SRE)
Data Science & AI Engineering	Make sense of all kinds of information and teach computers to make helpful guesses and predictions.	Data Scientist, Machine Learning Engineer, Data, Engineer.
Cybersecurity & Threat Defense	protect computer networks, check for weak spots, and fix things quickly if something goes wrong or dangerous happens	Information Security Analyst, Penetration Tester, SOC Specialist.
Information Systems & Enterprise IT	Match how people work with the right computer tools and big business software setups.	Business system analyst, IT manager, Solution Architech.

My Chosen Pathway (Part B)

I am most interested in:

Why I chose it:

I am most interested in front-end so it's inside the Software Engineer. I like to explore designing the layouts and anything that people see whenever they open a website.

I chose it because ever since I learned programming, building the outer part of a website caught my attention. That's why I know I'm interested in this and I want to improve these skills.

Table 2. My Career Roadmap for My Chosen Pathway (Part C)

Career Stage	Example Job Title(s)	Key Skills / Responsibilities
Entry-Level	Junior Front-End Developer Front-End Developer I Web Developer (Junior)	Solid foundation in HTML5, CSS3, JAVASCRIPT. Knows how to fix a minor bug and maintain existing code. Familiarity with basic tools such as VS code.
Mid-Level	Front-End Developer Front-End Engineer UI Developer	Work with design systems & component libraries - Mentor junior developers & lead small features
Senior-Level	Senior Front-End Developer Lead Front-End Engineer Senior UI/UX Engineer	Evaluate & adopt new tools/technologies Establish coding standards, style guides, & code review processes Design scalable front-end architecture & project structure
Expert / Leadership	Front-End Architect Principal Engineer Head of Front-End / Technical Lead	Oversee multi-team projects & ensure technical consistency Make high-level decisions on tech stack, infrastructure, & tooling Drive innovation, research emerging technologies

Table 3. Skill Gap Self-Assessment (Part D)

Essential Competency	Already Have It? (Y/N)	How I Plan to Learn It
Mastering HTML5 semantics and structures	N	I plan to study more about this, and practice code.
Debugging & browser devtools	N	Practice more using my own code and watch some tutorials
Designing System and UI	N	Study more about UI so I could improve myself.
Accessibility	N	The accessibility of coding is something that hard for a student like me.
Basic Performance	y	I already know to do some basic even just a little

Essential Competency	Already Have It? (Y/N)	How I Plan to Learn It
React core	N	I'm planning on how to react when something become or turned to error.

Table 4. Real Job Posting Found (Part E, Optional / Bonus)

Item	Details
Job Title	Junior Front-end Developer
Company	Internex.io
Platform / Source (e.g., LinkedIn, Indeed, JobStreet)	Linkedin
Required Skills Listed (3–5)	Strong fundamentals in HTML, CSS and JavaScript Basic understanding of responsive web design Building responsive, cross-browser web apps

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Why did you choose this particular pathway over the other four?

I choose the front-end or the pathway of software engineer because among the others, this is what I wanted to be. Building a website or application designs is what I always wanted. I am not so good at this right now, but I know, everything can be learned and there is no impossible.

2. Looking at your roadmap, which career stage excites you the most, and why?

The entry level, because I believe everything is hard at the first time. It excites me because there I will experience different kinds of circumstances that could make me grow into a bigger person. The first time might be the scariest, but it is also the most exciting and full of learning.

3. Which required skill or competency for this pathway do you already have, and which one will be hardest for you to develop?

I already have the skill in basics of HTML and CSS, but I still need improvements. Knowing the things doesn't make us already done. In technology we must learn more and earn more knowledge. But the thing that I think would be hard for me is the adopting new tools, because not in every way we could learn something new that easily, so maybe, it's the skill that would be hard for me to develop.

4. How does mapping this roadmap change — or confirm — your plans for the rest of this course and your degree?

This activity makes me realize the things that I really wanted and why I choose to study the complexity of technology—it's because I want to build a website or application that could help the community. Its like a small guide to visualize what I really wanted after pursuing computer science.

VII. Scoring Guide

Criteria	Description	Points
Completeness	Tables 1–3 and the reflection are fully filled out (Table 4 is bonus).	25
Roadmap Accuracy & Depth	Roadmap and skills recorded match the lecture pathway (or credible outside research for IS & Enterprise IT).	25
Realization / Reflection	Answers are thoughtful, specific to the student's own choice, and well written.	30
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	20
Total		**100**

Checked by: **Score:** / 100

Instructor's Signature over Printed Name / Date